Using Virtual Reality to Elicit the PLAY Emotion

Devin Marais
University of Cape Town
Cape Town, South Africa
mrsdev003@myuct.ac.za

Abstract

Emotions are complex, yet highly informative aspects of human cognition. The Discrete and Dimensional Models of emotion are the most commonly accepted frameworks today. The Discrete Model poses that emotions are distinct classes, whereas the Dimensional Model defines emotions according to combinations of valence, arousal and dominance. The Theory of Affective Neuroscience acts as an alternative to these cortical models, and classifies emotions according to the evolutionary, subcortical systems in the mammalian brain. PLAY is an emotion from the Theory of Affective Neuroscience and relates to social play. PLAY has significant impacts on one's psycho-social development and functional capacities. Studying emotions is done through controlled emotion elicitation experiments, using either passive or active emotion elicitation techniques. The active techniques provide greater ecological validity making them preferable over passive methods. The emotions elicited are measured through subjective questionnaires or objective physiological measures. Virtual reality (VR) is an effective active emotion elicitation technique, specifically due to its ability in evoking a sense of presence. The intensity of the emotions elicited depend on the presence felt in the VR. The presence induced in a VR depends on its level of immersion, interactivity, affective content and believability/realism. Eliciting a specific emotion in VR requires selecting design components with valence, arousal and dominance ratings corresponding to those of the target emotion. These design components include semantic, sensory, dynamic and interactive elements.

1 Introduction

Virtual reality (VR) has recently emerged as a powerful tool for emotion elicitation. Its well-documented success has been attributed to the strong sense of presence it can induce via its immersive and interactive nature. While the current literature covers a wide variety of emotions being elicited through VR, despite its heavy implications in social and cognitive development, there is a large gap in eliciting the PLAY emotion, defined under Jaak Panksepp's Theory of Affective Neuroscience, through VR. This literature review aims to investigate how VR may be used to elicit PLAY, by exploring different emotion classification frameworks, elicitation techniques and experimental design procedures for affective VR's.

2 Theoretical Frameworks of Emotions

Emotions are core to the human experience, influencing decision making, behaviour and cognition [37, 45]. From this, it is understandable that one would want to formalise emotions such that they may be studied and isolated in a more controlled manner. This notion of formalising emotions has been a topic of debate for more than a century [16], during which several emotion classification frameworks were developed. Of these, the Discrete Model of Emotions [15] and the Dimensional Model of Emotions [41] have proven to be most widely accepted. In the Discrete Model, human emotions are categorised according to a set of distinct emotional states [15], whereas the Dimensional Model posits that emotions lie on a three-dimensional plane of arousal (excitement), valence (pleasantness) and dominance (control or influence) [41].

These two models were developed using interpretations of human facial expressions, behaviours and self-reported feelings. Hence, they were based on cortical (conscious) human processes [15, 16]. As an alternative to these models, Jaak Panksepp's Theory of Affective Neuroscience was created based on the unconscious, evolutionary cognitive systems he observed in all mammals [34]. Panksepp [34] used subcortical processes in his framework for emotions, and identified seven primary emotions: SEEKING, LUST, CARE, FEAR, RAGE/ANGER, PANIC/SADNESS and PLAY.

3 PLAY as an Emotion

Panksepp et al. [35] grew the body of work on PLAY using experiments conducted on rats. During these experiments, he identified the act of rats pinning each other down to be the signifier of PLAY [35]. This pinning was not an aggressive action, as would occur during physical altercations, but was a non-serious expression of social play [35]. During these bouts of pinning, the rats would run around joyously and switch between being pinned and doing the pinning [35]. From these observations, Panksepp et al. [35] defined PLAY as a non-serious, highly-pleasurable sequence of back and forth interactions between entities.

PLAY, or more generally social play, has significant implications for the development of a rich social repertoire [47]. Moreover, during PLAY, one's social, cognitive and motor skills are formed and fine-tuned for adaptive environments [47]. Aberrant social play during one's formative years has shown to be correlated to an array of psychiatric disorders such as autism, attention deficit disorder and early-onset schizophrenia [2]. Above these psychiatric disorders, dysfunctional or inadequate social play is linked to cognitive impairments and an inability to adapt in unpredictable or challenging environments [1].

In essence, sufficient PLAY, in quantity and quality, is imperative for one's functional capacities and overall well being. While PLAY is unique to Panksepp's theory of Affective Neuroscience, it can be mapped to the contemporary models of emotion classification. Considering the scarcity of PLAY's consideration in modern literature, and the abundance of that for the contemporary models, mapping PLAY to those models will enable this literature review to cover significantly more published work. This mapping will require us to decompose PLAY into its constituent components based on its

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definition. PLAY is an energetic, back and forth, highly pleasurable social interaction. This energetic component maps to high arousal, the pleasurable aspect speaks to a high valence and the back and forth nature equates to high dominance [41]. This high dominance is justified as the back and forth nature of PLAY requires actors to exert influence over their environment and respond accordingly to moves made by other players. This constitutes as having high influence, hence high dominance. Therefore, according to the Dimensional Model, PLAY is classified as a high arousal, high valence and high dominance emotion. Whereas for the discrete model, PLAY sits as an emotion adjacent to joy and surprise [15]. Considering the importance of PLAY for human development, further investigation of its inner workings and methods via which acceptable PLAY may be facilitated is warranted. Therefore, eliciting PLAY in a laboratory environment where it may be studied more closely is deserving of more attention.

4 Emotion Elicitation

Emotions are complex systems which have proven to be difficult to define, making them particularly challenging to study. Despite this, considering the fact that emotions play crucial roles in human creativity, cognition and behaviour [45], there is much value in deepening the current understanding of these systems. The scientific community has acknowledged this, and has developed techniques via which emotions may be more reliably studied. Of these techniques, eliciting emotions in a laboratory controlled environment is regarded as the standard [16].

Emotion elicitation involves inducing a particular emotion in a test subject via some mechanism, such that their response (i.e. the emotion) may be observed or analysed in greater detail [16]. Two main categories of emotion elicitation techniques have emerged, passive and active techniques. Passive techniques involve the subject observing emotional content. Examples include watching videos, observing images, recalling past experiences and listening to audio [16]. Active techniques entail the subject actively engaging in the emotional experience. Examples include interacting with humans or avatars, virtual reality and games [16, 45]. While the traditional, passive methods have been able to successfully elicit emotions, their lack of ecological validity (i.e. degree to which the findings can be generalised to real-world settings) and immersion [46] limits the intensity and real-world relevance of the emotions. Contrastingly the contemporary, active emotion elicitation techniques facilitate greater immersion and subjectivity, which results in more intense and emotional experiences [46]. Among the active methods, virtual reality has gained significant traction over the past years and is promising as a strong emotion elicitation medium for future studies [45].

5 VR as a Tool for Emotion Elicitation

5.1 The Emergence of VR in Emotion Elicitation

In recent years, virtual reality (VR) has gained traction as an effective active emotion elicitation tool. The success of VR in eliciting emotions is most commonly attributed to its ability to invoke a strong sense of presence, which is defined as the sense of "being there" [40]. It has been shown that the level of presence felt is correlated with the emotional intensity felt by the subject [4, 40]. VR has been able to elicit a wide array of emotions, across both the Discrete and Dimensional Models of Emotion. Riva et al. [40] found that through virtual parks, they were able to elicit both anxiety and relaxation in their test subjects. Chirico et al. [8] was able to induce the emotion of awe using 3 separate virtual environments (VE's). Felnhofer et al. [12] developed 5 different virtual parks, each aimed at eliciting different emotions, and succeeded with four out of these five parks. The efficacy of VR in emotion elicitation has been well documented in the current literature [45], where the aforementioned examples are but just a few instances of its success. However, the experimental procedure for implementing VR for emotion elicitation has not yet been formalised to one specific framework.

5.2 Implementing Emotion Elicitation in VR

During the experimental design of an emotionally-inductive VR, both the target emotion and evaluation criteria must be clearly defined. In most of the current literature covering VR's usage in emotion elicitation, the target emotions are defined according to the Dimensional Model of Emotions. For example, [47] defined joy as being 0.76 valence, 0.48 arousal and 0.35 dominance whereas fear was defined as -0.64 valence, 0.60 arousal and -0.43 dominance (ranging from +1 to -1). With the dimensions defined, one can begin selecting design features that align with the desired outcome. As important as defining the goal, we must also define our evaluation criteria. Since our goal is defined in terms of desired valence, arousal and dominance, our evaluation must determine the actual valence, arousal and dominance experienced by the subjects.

In most current VR emotion elicitation studies, both subjective measures (questionnaires) and objective measures (physiological responses) are used to determine the actual scores experienced in terms of these dimensions [45]. The subjective measures include several questionnaires which test subjects complete following the emotional experience. These questionnaires, while long-standing and well-used, rely on subjective reports which may be skewed due to cognitive bias, forgetfulness of what occurred during the experience and misinterpretation of the questions posed [33, 45]. Furthermore, this measure does not allow for continuous monitoring of the subject's responses during the entire emotional experience. This means that potentially useful information is being lost via only gathering data at the end of the experiment. As a solution to the limitations of subjective measures, objective measures provide more reliable figures with which the success of the experience may be evaluated. Of these questionnaires available, the Valence-Arousal-Dominance questionnaire (VAD) and the Positive-And-Negative-Affect-Schedule (PANAS) have been used frequently [25, 45]. Objective measures make use of physiological signals, measured using apparatus during the experience, which correlate to specific scores across the three dimensions. Of the many physiological signals, electroencephalography (EEG) data, heart rate variability (HRV) and galvanic skin response (GSR) exist as the most extensively employed measures [37]. EEG works by measuring the electrical signals of the brain and mapping it to brain waves [7]. As an example of this potential usage, gamma brain waves are associated with higher levels of arousal [7], and high valence emotions are associated with increased frontal alpha and right posterior

beta power [37]. HRV has been proven as a strong indicator of autonomic nervous system responses to positive and negative stimuli [37] making it a suitable indicator for arousal. GSR measures the activation of the eccrine sweat glands, which activate according to the body's fight or flight response [7] and is sensitive to high arousal emotions [37].

It is important to note, however, that while the objective measures may be more accurate they struggle in measuring valence [7] therefore subjective questionnaires, which can capture valence better [45], should still be used in an affective VR implementation. Table 1 provides a summary of the most frequently employed subjective and objective measures, which is based on three separate meta-reviews for VR based emotion elicitation [7, 25, 45].

5.3 Designing a VR for Emotion Elicitation

The efficacy of VR in eliciting emotions has been widely attributed to its ability to evoke a strong sense of presence [8, 40, 45]. Felnhofer et al. [12] went as far as saying that presence is a prerequisite for a VR to elicit any emotions at all. Additionally, the success in evoking the target emotion requires us to elicit its corresponding levels of valence, arousal and dominance which is determined by the individual design components. Therefore, the design process for an affective VR must consider both presence and design elements.

5.3.1 Presence as a Design Consideration. Presence is most commonly defined as the sense of "being there" [40], where the subjects feel as though the VR is real life. However, as pointed out by Baños et al. [4] presence is not a monolith. It is driven by multiple components. Immersion and interactivity are two of these components [4] which expands our definition to "being there, doing there". Furthermore, Riva et al. [40] discovered that presence is heavily influenced by subjects' emotional reactions, which is informed by the affective content presented in the VR. Lastly, the extent to which the subject consciously believes they could feasibly exist in the VR's scenario impacts the level of presence felt [9, 13, 27, 39]. This culminates our final definition of presence to "being there, doing there, feeling there, believing there". To maximise the presence induced by a VR, these individual components must be optimised.

Immersion speaks to how isolated the subjects feel from the outside world in the VR experience [8]. This level of isolation is mostly related to the technology used in the experience [4] which could be fully-immersive (e.g., head-mounted displays) or non-immersive (e.g., computer screens). Baños et al. [4] used two virtual parks to study the relationship between immersion and presence. One park was designed to elicit sadness, while the other was designed to elicit a neutral emotion. Participants were split into three groups and assigned to use equipment with differing levels of immersion. The equipment consisted of head-mounted displays (HMD), rear-projected videos or PC-monitors - stated in decreasing levels of immersion. The results indicated that the more immersive technologies produced greater levels of presence. Hence, it is suggested that immersion and presence are positively correlated.

Interactivity refers to the subject's ability to influence, control or alter the VR [10]. Examples of interactivity include subjects being able to control which parts of the VR they explore and having the ability to move objects in the VR. In the VE's developed by Chirico et al. [8] aimed at eliciting awe, they allowed subjects to navigate through the VE on their own accord which provided greater interactivity. This study indicated that stronger levels of interactivity provided greater presence. Moreover, Madsen [24] found stronger expressions of fear in subjects when they were granted more control over the VR experience. In this study, subjects were presented with a horror scene. One group only watched the horror scene, while the other was given controllers to play the horror scene as a game. In the controller condition, the actions of the subjects determined the scene's events and provided greater interactivity. This finding proposes that greater levels of interactivity produce stronger emotional responses.

Affective content refers to the emotional material of the VR, aimed at inducing emotional reactions. Hidaka et al. [17] placed subjects in a VE displaying a man burying a coffin in the desert, with audible sobbing. The aim of this VE was to elicit distress, and this man acted as the affective content. In the fear-inducing VE developed by Liao et al. [22], they used the setting of an abandoned, dark house as the affective content. Additionally, Gilpin et al. [14] used the affective content of a pet dying in their VE aimed at eliciting sadness.

The affective content employed in a VR determines the emotion induced, but more importantly, its quality determines the strength of the emotional response [45]. To explain, compare the affective content of receiving a free ice cream versus that of winning the lottery. While both may elicit happiness, it is reasonable to expect the lottery winning content to produce stronger expressions of joy. With this, we could say the lottery affective content was of a higher quality than that of the ice cream content.

This relationship between the quality of affective content and presence is nuanced. Riva et al. [40] placed subjects in one of two virtual park conditions, where one was emotionally-charged and the other was non-emotionally charged. They found that subjects experienced higher levels of presence in the emotionally-charged park. However, they concluded that this relationship between emotional intensity and presence was bidirectional. Felnhofer et al. [12] later built on top of this using their five virtual parks, all aimed at eliciting different emotions. They found that the levels of presence reported across all parks were similar, implying that the higher presence observed by Riva et al. [40] was due to the inclusion of affective content and not related to the emotion elicited itself.

Using this, we can say that since the quality of the affective content informs the strength of the emotional response, and the strength of the emotional response informs the level of presence, the quality of the affective content influences the presence felt.

Similarly, believability and realism play a role in the presence felt during a VR experience [9, 13, 27, 39]. Believability is the extent to which the subject believes the VR is real life, whereas realism speaks to how accurate the VR's details are to the physical world. While slightly different, believability and realism are intertwined. Without a realistic VR the subjects would not believe the VR is real life, and if the content presented in the VR is not believable (e.g., a fairy-tale land) the benefits from accurate VR details would be negated. It has been suggested that higher levels of believability and realism promote a stronger sense of presence [9, 13, 27, 39].

The impact of believability was apparent in the awe-inducing VEs developed by Chirico et al. [8]. One of these VE's presented subjects with a view of the Earth from outer-space. The levels of

Table 1: Summary Table of Commonly Used Subjective and Physiological Measures

Category	Measure	Capture Capabilities	Strengths	Limitations	Source
Subjective	Valence-Arousal-Dominance Questionnaire	Valence, arousal and dominance	Applies directly to Dimensional Emotion Model; is simple	Relies on self-reports which can be biased	[25, 45]
Subjective	Positive-And-Negative-Affect-Schedule	Positive and negative emotions	Long-standing and validated; widely used; measures valence well	Not directly tied to arousal or dominance	[7]
Subjective	Discrete emotion labels	Specific discrete emotions	Intuitive and easy to use	Lacks precision	[45]
Physiological	Galvanic Skin Response	Arousal	Highly sensitive to arousal; measures arousal well	Lacks ability to measure valence and dominance	[25, 45]
Physiological	Heart Rate Variability	Arousal	Reliable measure of physiological activation	Can be influenced by non-emotional factors	[25, 45]
Physiological	Electroencephalography	Brain activity associated with emotion	Provides highly useful neural data	Complex setup; less practical for VR experiments	[7]

presence reported for this VE were lower than those for the neutral VE. The authors attributed this low presence to subjects consciously knowing they could not feasibly be in outer-space. In other words, the subjects did not believe the VE and as such, they felt lower levels of presence.

Additionally, realism can be enhanced through multi-sensory and physically engaging VR's [19]. This aligns with the findings of Slater et al. [44] and Slater and Steed [43]. Slater et al. [44] placed subjects in VE's containing trees of varying heights, using HMD's. In VE's with higher trees, subjects had to move their heads further back to see the tree's top, which constitutes as greater levels of physical engagement. It was found that higher levels of presence were experienced by subjects in VE conditions with higher trees. Therefore, it was concluded that there is a correlation between the degree of bodily engagement and presence in a VR. Slater and Steed [43] presented subjects with a VE in which they had to move chess pieces. One group had to use their hands to reach out and move the pieces, whereas the other used a mouse. The group using their hands exerted more physical effort than the mouse condition. The group using more physical engagement reported greater levels of presence, which again suggests that bodily involvement and presence are correlated.

Conversely, Aymerich-Franch [3] found that no correlation between bodily involvement and presence exists. Subjects were presented with a VE, in which they had to complete some arbitrary task. They were randomly separated into two groups. One group had to use their entire body to complete the task, whereas the other used a joystick. No significant differences in presence were reported between the two conditions. However, in this study only three out of seven groups in the full-body usage condition completed the VE's task, whereas all groups in the joystick condition completed it. As pointed out by Dozio et al. [10], when presenting users with any task in VR, they must be motivated to complete it, or else they disengage. This act of disengaging reduces subjects' interaction with the VR and subsequently reduces presence [8]. Considering that majority of the participants in the body condition did not finish the task, it could be argued that they lost motivation during the experience and disengaged. Furthermore, in this study several participants reported feelings of frustration during the experience due to lags occurring in the VE. It is reasonable to say that during these lags the subjects were reminded that the VE was not real life as lags do not occur in the physical world. Hence, it could be argued that the lower levels of presence observed stemmed from subjects disengaging during the task, and being reminded that the VE is not real life due to the system lags. With this, the notion that increasing bodily engagement in a VR to increase realism and presence still stands.

Thus far we have discussed the methods to optimise presence, which is summarised in Figure 1. While this optimisation is imperative, adjusting the VR to elicit the specific levels of valence, arousal and dominance associated with the target emotion is just as important. This modulation is achieved through the selection of individual design elements.

5.3.2 Design Elements for Affective Virtual Reality. Much like any experimental procedure, the experience with which the subjects are presented must be curated to produce the desired outcome. In the context of affective VR, this refers to selecting design elements to elicit the levels of arousal, valence and dominance associated with the target emotion. Dozio et al. [10] proposes a four-component approach to designing an affective VR experience. These components include semantic elements, sensory elements, dynamism and interactivity. While the latter three components can be developed

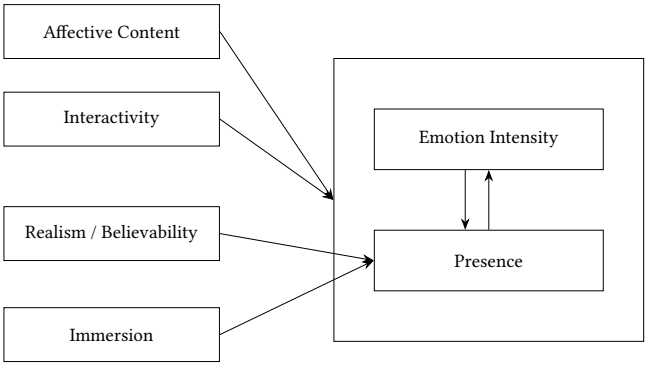


Figure 1: Conceptual model of factors influencing presence and emotion intensity.

in parallel, the semantic elements must be established first as they cement the basis on which the other components build.

Semantic elements provide the overall meaning and context of the VR. Deriving such elements requires identifying themes generally associated with the target emotion [10]. For example, joy is associated with fun, celebration and childhood. Once these elements have been defined, the rest of the design process aims to communicate them across to the subject.

These semantic elements carry concrete associations, which inform the general context of the VR design. For example, the semantic element of vacation is associated with the beach and clear, blue skies. With this, a VR aimed at communicating vacation would set the scene for the VR to include a beach on a sunny day. With this groundwork laid, the next step is to fine-tune the VR such that it elicits the targeted levels of valence, arousal and dominance. This editing process is done by adding the sensory, dynamic and interactive elements.

Sensory elements refer to those which stimulate the human senses. Under the framework proposed by Dozio et al. [10] we will focus on visual, auditory and olfactory senses.

These visual aspects include both concrete objects (e.g. human figures and vegetation) and abstract elements (e.g. colour and lighting). For concrete objects, the International Affective Picture System (IAPS) provides a set of images with pre-labeled valence and arousal levels [6]. These ratings can be extended to apply to VR implementations of the images. As an example, Dozio et al. [10] aimed to elicit happiness in one of their VE's. Happiness has both a valence and an arousal with ratings between 5 and 9, where 1 is the minimum and 9 is the maximum [7]. They created a VE containing the following concrete visual components: a beach, hot-air balloons and the sea. According to the IAPS these components have valence scores of 8.22, 6.97 and 8.03 along with arousals of 5.71, 4.98 and 5.46 respectively [6]. From this, we can see that the visual elements included had valence and arousal ratings corresponding to those of the target emotion. Similarly, Jiang et al. [18] included vegetation in their VE "Jetty on a Lake" which aimed to elicit high valence (between 5 and 9) and low arousal (between 1 and 5). The IAPS has labelled vegetation with a valence of 7.06 and an arousal of 3.83 [6]. Therefore, the visual components lined up with the targeted valence and arousal levels.

Abstract visual elements also impact the levels of valence, arousal and dominance elicited in a VR [10]. Naz et al. [33] studied the effects of colour, light and texture by modifying these parameters in a VE consisting of a desk, a chair and a bed. Subjects were exposed to various combinations of these parameters where colour moved from cool to warm, light changing from high to low and texture ranging from smooth to rough. They found that warmer colours evoked greater levels of arousal than the cooler colours, irrespective of the brightness levels. This is in agreement with the findings of Matthias and Beckhaus [26] whereby subjects were exposed to thirty different Means of Expressions (i.e. small adjustments in various parameters) implemented in an affective VE. They found that red (warm) colours produced higher levels of arousal than blue (cool) colours. Moghimi et al. [28] conducted a similar study where the parameters of their VE, in which subjects controlled a speedboat on open waters, were slightly adjusted such as colour and blur. They found that lower levels of valence were elicited when the VE was presented in black-and-white as opposed to in-colour. This can be generalised to say that the use of an achromatic colour scheme produced lower levels of valence in this study. This draws a parallel to the virtual park aimed at eliciting boredom developed by Felnhofer et al. [12]. Boredom is defined as a low valence and a low arousal emotion [32]. In this virtual park, an achromatic colour scheme was employed, hence the colour choices produced lower levels of valence aligning with those of the target emotion, being boredom.

In the same study mentioned above, Naz et al. [33] found that brightness impacted perceived spaciousness, where higher brightness made the VE seem more spacious and lower brightness resulted in the VE feeling smaller. While the impact of spaciousness on valence, arousal or dominance was not considered in this study, Presti et al. [38] found that tightly enclosed spaces increased levels of arousal, hence for very low levels of brightness in a VE we can reasonably say that arousal would be increased. Furthermore, smaller spaces may feel bounding meaning that subjects are have less freedom in the VE. Hence, we can say that low brightness levels would reduce perceived freedom and decrease levels of dominance. Lastly, Naz et al. [33] noted that the general aesthetics of the virtual space impacted valence, where the more aesthetically pleasing the space was, the higher the valence levels were and the opposite is true too. This claim has also been made by Li et al. [21] where they drew from the principles of environmental psychology.

As a continuation to our discussion on sensory elements, auditory aspects have been shown to modulate levels of valence, arousal and dominance [10, 26, 29]. Much like visual components, sounds can be either concrete or abstract [10]. Concrete sounds are those which occur naturally in the real world, such as the sounds of ocean waves or human screams. Conversely, abstract sounds are manufactured which includes music and melodies. In terms of concrete sounds, the creators of the IAPS developed a similar database for sounds in the International Affective Digitised Sounds system (IADS) [5]. The IADS lists various concrete sounds with their respective valence and arousal levels attached [5]. With this, sounds can be selected based on their valence and arousal levels such that they align with those of the target emotion. For example, Dozio et al. [10] used the sound of a woman screaming in their VE aimed at eliciting fear. According to the IADS, this sound has a valence of

1.93 and an arousal of 7.77 where the minimum score is 1 and the maximum is 9 [5]. Fear is categorised as a low valence (between 1 and 5) and a high arousal (between 5 and 9) emotion. From this, we can see that the sounds used in this VE matched the target emotion according to its dimensional ratings. Similarly, the joy-eliciting VE developed by Felnhofer et al. [12] included the sounds of twittering birds which has been shown to evoke high levels of valence [26] which aligns with the high valence score of joy [7].

In terms of abstract sound, both pace and tone have been found to impact valence, arousal and dominance [11, 26, 29]. Droit-Volet et al. [11] found that the faster the rhythm of the music, the more arousing it was. This aligns with the findings from the numerous Means of Expression designed by Matthias and Beckhaus [26], where they concluded that when subjects were exposed to suspense music, which has a faster pace, they experienced higher levels of arousal. Moreover, tonal (harmonious) sounds produced higher levels of valence, whereas atonal (chaotic) sounds elicited lower levels of valence. Molet et al. [29] had similar findings whereby they placed subjects into a VR house with either consonant (harmonious), static or dissonant (chaotic) music playing. They found that subjects experienced the lowest levels of valence in the dissonant condition, and the highest levels in the consonant condition.

To end our discussion on sensory components, we will consider olfactory elements which are related to smell. Unlike visual and auditory elements, smell does not directly manipulate the levels of valence, arousal and dominance. Rather, it is able to aid the other design elements through its ability to enhance presence [31]. Munyan et al. [31] placed subjects into a mildly anxiety-inducing VE where the condition of scent or no scent was employed. They found that higher levels of presence were experienced for the scent condition, hence indicating that including olfactory stimulants in an affective VR may increase presence. This was, however, contested by Serrano et al. [42]. They conducted an experiment whereby subjects were placed into a VE aimed at eliciting relaxation, and the conditions of no scent or touch, only scent, only touch and scent with touch were used. They did not find significant differences in presence between all four conditions, suggesting that there is no correlation between smell and presence. However, the authors noted that the subjects had high baseline levels of relaxation prior to the experiment being conducted. It could be argued that with this high baseline, the subjects more easily reached the maximum intensity of relaxation across all conditions. Since emotional intensity and presence are positively correlated [40], if emotional intensity were to reach its limit it would be expected for presence to reach its maximum as well. Therefore, diminishing marginal returns for increases in presence gained from the inclusion of scents could have possibly occurred in this study. Therefore, the benefits in presence from olfactory stimulation would not have been evident. Moreover, it has been found that there is an association between arousal and presence, where high arousal emotions provide higher levels of presence [8]. Since relaxation is a very low arousal emotion [42], increases in relaxation would lower arousal and as such, lower presence. This would have counteracted the increases in presence due to the inclusion of scent, which could explain why Serrano et al. [42] did not observe a correlation between scent and presence.

As a third consideration in the design framework developed by Dozio et al. [10] dynamism refers to how the VR changes or moves over time. This manifests in two ways: through the movement and evolution of the VR and its constituent components or through the subject being able to choose which part of the VR they are observing. The former includes the movement of concrete objects in the VR and is closely related to arousal, whereas the latter speaks to how restricted the subject is with respect to their allowed actions in the VR and is linked to dominance [10]. Faster moving dynamics have been shown to increase arousal [36]. To study this relationship further, Mousas et al. [30] created two virtual avatars, one of a human male and the other of a zombie male. Participants were placed in a VR with these avatars, separately, and the speed of the avatars' movements were changed from slow to fast. They found that for both avatars, the faster moving conditions evoked higher levels of arousal and lower levels of valence. However, the lower valence scores reported may not have been due to the dynamics of the VR. Zombies are associated with fearful emotions and are threatening figures. For the human male, it is possible that the uncanny valley effect occurred, whereby human figures with low realism produce cold and fearful emotions in observers [23]. It could be argued that when these already frightening figures began to move faster, the level of fear induced in the subjects increased. Since fear is a low valence emotion [7], increases in fear would have caused lower valence scores which could explain the lower valence levels reported by Mousas et al. [30].

Lastly, interactivity as a design component speaks to the influence and the level of control the subject has over the VR. While interactivity plays a substantial role in presence, as discussed in section 5.3.1, the fact that it relates to control implicates it as a modulator for dominance. It has been found that the more control a subject feels they have in a VR, the higher their levels of dominance are [24]. Therefore, by increasing interactivity within an affective VR, we can increase levels of dominance.

In this section, we have explored how sensory, dynamic and interactive elements alter the specific levels of valence, arousal and dominance in the subjects. For clarity purposes, a summary of this is shown in Table 2. With these different levers defined, creating an affective VR becomes a "plug-and-play" system whereby once the valence, arousal and dominance ratings of an emotion are, we can simply choose elements which correspond to these levels.

6 Designing Affective VR for Eliciting PLAY

Up until this section, we have discussed the general VR emotion elicitation methods, the experimental design procedure and affective VR design principles. We will now integrate these topics and curate them for eliciting the PLAY emotion specifically. To start, we will expand upon our definition of PLAY under its Dimensional Model mapping and Discrete Model adjacents.

In section 3, we defined PLAY to be high valence, high arousal and high dominance. To assign specific values (on a range from 1 to 9) to what we mean by "high", we will use those assigned to PLAY's discrete adjacent emotions, being joy and surprise, followed by any necessary adjustments based on PLAY's definition from section 2. The DEAP database assigned joy with a valence between 5 and 9, an arousal between 5 and 9 and a dominance between 4 and 7 [20]. This database has also assigned surprise with a valence between 5 and 9, an arousal between 5 and 9 and a dominance

Table 2: Summary table of design elements and associated influences across valence, arousal and dominance.

Design Element	Valence	Arousal	Dominance
Sensory			
Visual			
colour brightness	✓	✓	
colour warmth	✓	✓	
colour chromaticism	✓	✓	✓
lighting	✓	✓	✓
aesthetics	✓		
Auditory			
pace		✓	
tone	✓		
Olfactory	✓	✓	
Dynamics			
speed		✓	
Interactivity			✓

between 4 and 5 [20]. Using these values, we can average PLAY's valence to 7, arousal 7 and dominance to 5. Our definition of PLAY emphasised that it was a highly pleasurable and energetic emotion with continuous feedback from the objects or beings being PLAYed with. Our current rating matches the highly pleasurable and energetic component of our definition. However, a dominance of 4.5 is too low for PLAY. Since PLAY is regarded as high dominance, we can adjust its dominance score to that of its valence and arousal. Hence, PLAY can be said to have a valence of 7, and arousal of 7 and a dominance of 7.

With this defined, we are able to follow the implementation procedure outlined in section 5 where we select our evaluation tools and design elements. The selection process is out of scope for this literature review and is rather included in the project proposal.

7 Conclusion

In this literature review we have discussed both contemporary models of emotion and Jaak Panksepp's Theory of Affective Neuroscience with a particular emphasis on his PLAY emotion and how it maps to the contemporary models. Following this we explored the different emotion elicitation techniques and found that active elicitation techniques are preferable over the passive techniques. Of these active techniques, virtual reality (VR) has emerged as a promising medium due to its ability to induce strong, ecologically valid emotions with its greater sense of presence. We formalised the experimental procedure for an affective VR into defining the target emotion, developing the evaluation criteria and designing the VR with the aim of optimising presence and including elements which elicit the valence, arousal and dominance associated with the target emotion. Through this, an affective VR for eliciting any emotion, including PLAY, can be successfully created.

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